

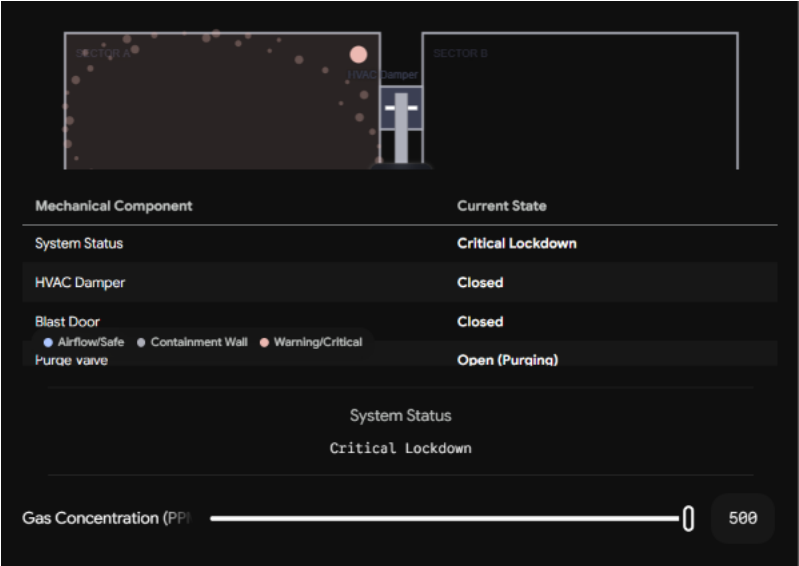


Fig 1.1 Critical Lockdown(Max)

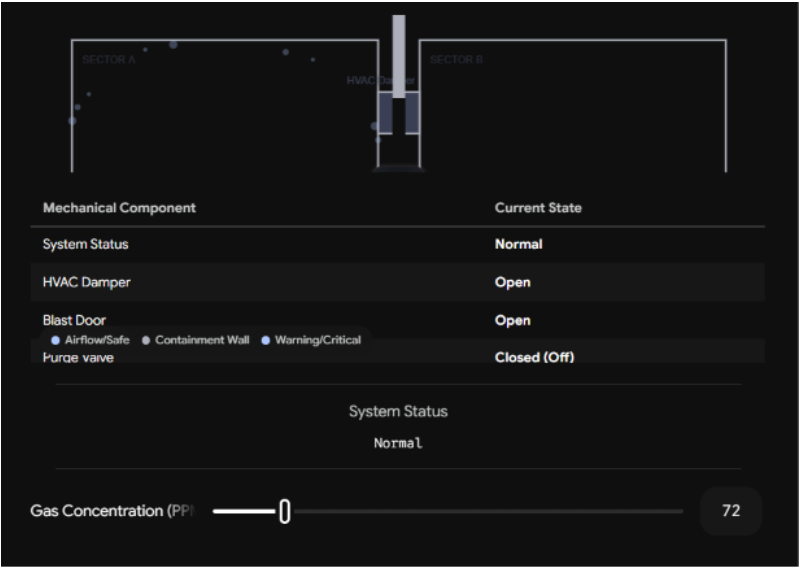


Fig 1.2 Normal

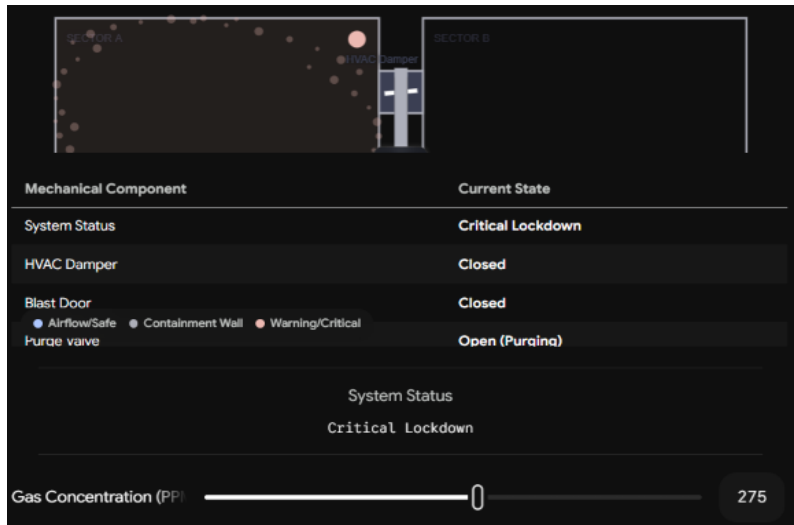
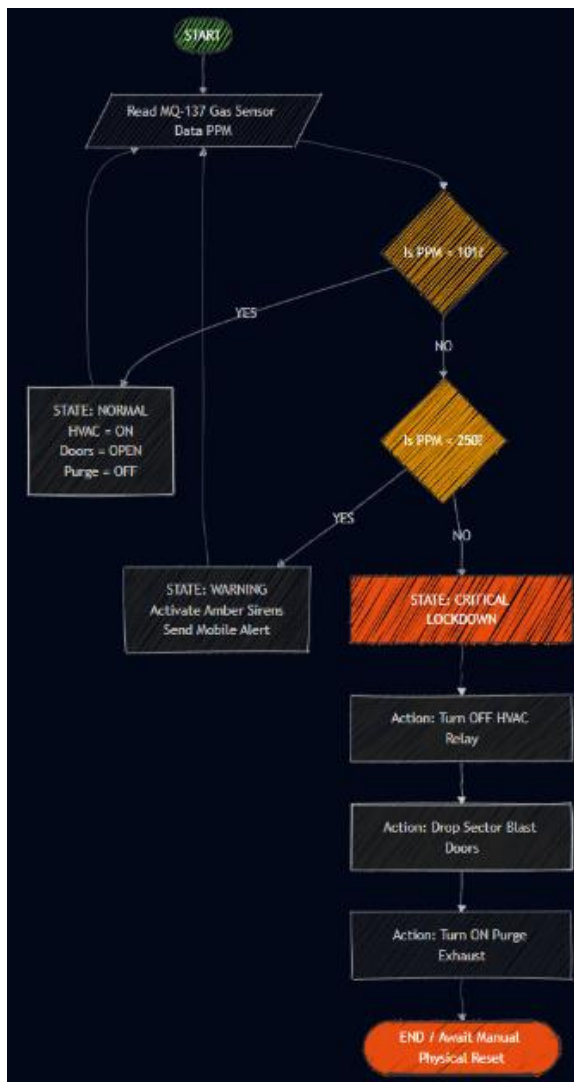
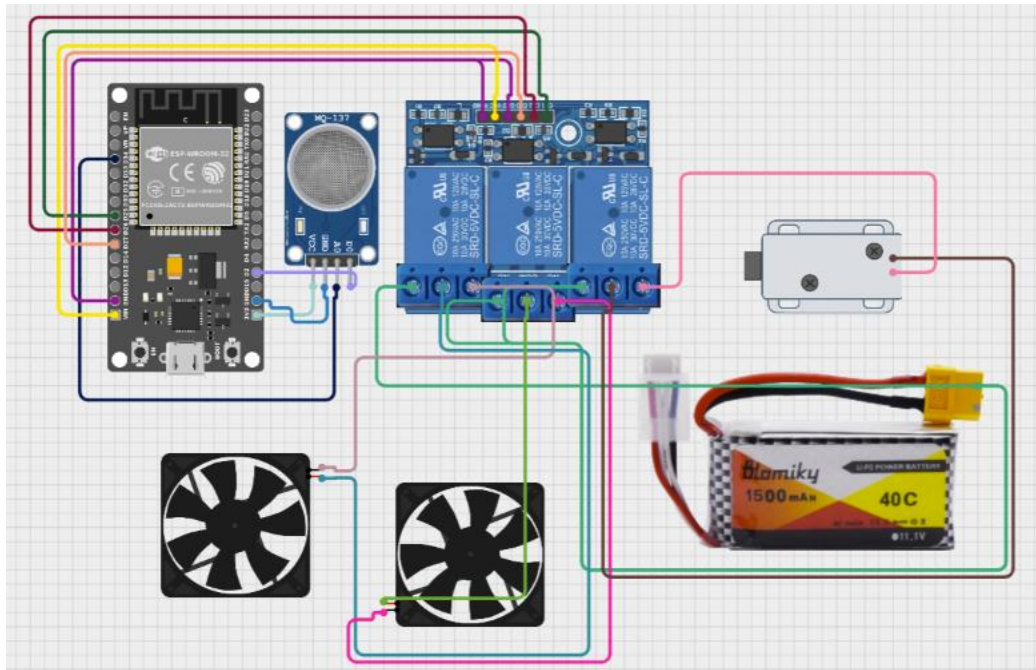


Fig 1.3 Critical Lockdown(Switch)





System Design Report: Automated Toxic Gas Containment & Purge System

1. Problem Statement & Objective

In industrial manufacturing and chemical processing, a toxic gas leak (such as ammonia, chlorine, or carbon monoxide) presents an immediate, life-threatening crisis. If a leak occurs, standard HVAC (Heating, Ventilation, and Air Conditioning) systems act as a distribution network, rapidly spreading the toxic gas to safe zones before human operators can react. The objective of this automation system is to design a localized, edge-computed state machine that detects hazardous gas concentrations and automatically executes a mechanical sector lockdown and air-purge sequence without requiring human intervention.

2. Hardware Architecture & Components

The system relies on a decentralized microcontroller communicating with industrial sensors and high-voltage relays to physically manipulate the environment.

- **Logic Controller (ESP32):** Acts as the central processing brain. It continuously polls analog data from the sensors and executes the programmed threshold logic on the edge, ensuring zero latency.
- **Gas Sensor (MQ-137):** Calibrated to detect specific industrial gases like ammonia. It outputs an analog voltage that the microcontroller maps to a Parts Per Million (PPM) value.
- **Opto-Isolated Relay Modules:** Used to safely switch high-voltage loads (like industrial exhaust fans and HVAC power lines) using the low-voltage 3.3V logic signals from the ESP32 microcontroller.
- **Actuators (Solenoid Locks / Actuators):** Attached to the factory blast doors and HVAC ventilation dampers to physically seal the room.
- **Emergency Purge Fan:** A heavy-duty, one-way exhaust system that routes toxic air through a chemical scrubber and safely outside the facility.

3. System Logic & Escalation Flow

The automation logic is built on a three-tier escalation protocol. This prevents false alarms from shutting down the factory while ensuring an immediate, automated response if a genuine leak occurs.

- **State 1: Normal Operations (0 - 100 PPM)**
 - *Logic:* The sensor reads trace amounts of gas, which is normal for an industrial environment.
 - *Action:* The microcontroller keeps the standard HVAC Relay **ON** (providing normal air circulation). The Door Actuators remain **OPEN**. The Emergency Purge Fan is **OFF**.

- **State 2: Warning Threshold (101 - 249 PPM)**
 - *Logic:* The sensor detects an abnormal spike in gas concentration. This indicates a potential minor leak or a buildup of fumes.
 - *Action:* The mechanical states (doors and HVAC) remain unchanged to avoid disrupting production. However, the microcontroller triggers an amber warning light relay and sends an automated MQTT (Message Queuing Telemetry Transport) payload to the facility supervisor to investigate the sector.
- **State 3: Critical Lockdown (250+ PPM)**
 - *Logic:* The gas concentration reaches a life-threatening level. The system enters a hard, mechanical override.
 - *Action:* 1. **Isolate:** The standard HVAC relay is cut to **OFF**, instantly stopping the spread of gas to other rooms. 2. **Seal:** The Solenoid Locks are triggered, mechanically sealing the hazardous sector's blast doors. 3. **Purge:** The Emergency Purge Fan relay is switched **ON**, creating negative pressure in the room to suck the toxic gas out and push it through the scrubbers. 4. **Alert:** Sirens are activated, and a critical evacuation alert is broadcast to the factory network.

4. Fail-Safe Considerations

To ensure absolute reliability, the system is designed using a "normally open/normally closed" safety principle for the high-voltage systems. If the microcontroller loses power or the sensor wire is completely severed, the relays will default to their unpowered state. This automatically cuts power to the solenoid locks (dropping the blast doors) and cuts the standard HVAC, ensuring that a broken system defaults to a safe, contained state.

5. Glossary of Technical Terms

- **Edge-Computing:** The practice of processing data locally at the source (on the microcontroller itself) rather than relying on a distant cloud server. This ensures immediate, zero-latency decision-making, which is critical for life-safety systems.
- **ESP32:** A powerful, low-cost microchip featuring dual-core processing and integrated Wi-Fi/Bluetooth capabilities, widely used as the central logic controller in modern IoT (Internet of Things) automation.

- **HVAC (Heating, Ventilation, and Air Conditioning):** The primary physical infrastructure used to move and filter air throughout a building. In a chemical crisis, an unsecured HVAC system acts as a dangerous distribution network for airborne toxins.
- **MQ-137:** A specific analog industrial gas sensor calibrated to detect ammonia and other hazardous vapors by measuring changes in the electrical resistance of its internal heating element.
- **MQTT (Message Queuing Telemetry Transport):** A lightweight, low-bandwidth messaging protocol optimized for IoT devices. It allows the microcontroller to instantly "publish" alert messages to a mobile dashboard.
- **Opto-Isolated Relay:** An electromechanical switch that uses a beam of light (internal LED and photodetector) to bridge the signal between the low-voltage brain (ESP32) and high-voltage appliances (fans). This prevents high-voltage electrical spikes from destroying the sensitive microcontroller.
- **PPM (Parts Per Million):** The standard unit of measurement for gas concentration, indicating the volume of toxic gas per one million volumes of ambient air.
- **Solenoid Lock:** An electromechanical locking mechanism. It contains a wire coil that acts as an electromagnet when powered, physically pulling or releasing a solid metal bolt to secure the blast doors.
- **State Machine:** A fundamental logic architecture in computer science where a system can only exist in one predefined "state" (e.g., Normal, Warning, or Lockdown) at any given time, transitioning between them only when specific conditions are met.